Did 2018 Tariffs Boost Manufacturing Growth? A Local Industry Approach

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Abstract

What were the impacts of the 2018 Trump tariff hike on manufacturing growth? Motivated by gravity theory, I develop an empirical strategy that exploits cross-state variation in pre-treatment sectoral import volumes, generating differential exposure to nationwide tariff increases. I find a $1 tariff-induced reduction in imports raises output by $0.63 at the state-sector level. Accounting for a number of spatial spillover and general equilibrium channels, I estimate an implied nationwide manufacturing output gain from import protection of $69.5 to $127.5 billion. However, I find a $1 retaliation-induced reduction in exports results in a $3.4 loss in output, roughly five times the per-dollar output gain from import protection. A back-of-the-envelope calculation implies an aggregate loss of $177.8 billion, suggesting export losses are larger than the gains from import protection.

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1 Introduction

> "We were at our richest from 1870 to 1913. That's when we were a tariff country" – Donald Trump (2025)

Can trade protection boost domestic manufacturing? In 2018, the United States broke from a nearly century-long trajectory toward trade liberalization that began under the Franklin Delano Roosevelt administration. Tariffs were imposed on nearly $300 billion of imports from rivals and allies alike, triggering a trade war and retaliatory tariffs on $127 billion of U.S. exports.

I develop a novel empirical design to study the effects of this episode. The key insight is that states differed sharply in how much they imported from each manufacturing sector before the tariffs. As the 2018 tariffs were national, these pre-existing differences generated differential exposure to the same sector-level tariff shock across states, a relationship that can be rationalized by a first-order approximation to a standard trade gravity equation. I exploit this within-sector, cross-state variation to estimate the effects of trade protection on domestic manufacturing activity.

For example, Washington state and Massachusetts had nearly identical GDPs in 2017, about $527 billion and $530 billion respectively, and populations of 7.4 million and 6.9 million. Yet in communications equipment manufacturing, Washington imported about $2.79 billion from China, compared with about $180 million in Massachusetts. This contrast is intuitive: Washington is located on the Pacific coast and is more geographically proximate to China, while Massachusetts is located on the Atlantic coast. Therefore, Washington was more exposed than Massachusetts when national tariffs on Chinese imports in this sector increased in 2018. In this sense, the tariffs offered more protection to Washington communications equipment producers than to otherwise comparable producers in Massachusetts, because Washington faced substantially greater pre-tariff import competition from China in that sector.[1]

[1] Some of this demand may spill over to producers in other states: for example, if Washington consumers

Methodologically, this design allows for an across-state, within-sector comparison, which is infeasible in popular within-country approaches. For example, a large empirical trade literature exploits cross-sector variation by comparing sectors that received higher tariff cuts to those that received smaller cuts after a nationwide trade liberalization episode (e.g. Pavcnik 2002, Khandelwal and Topaleva 2011, Fajgelbaum et al. 2020, Flaeen and Pierce 2024). However, sector-level trends correlated with tariffs – whether coincidental or targeted by policymakers – would be a concern for this design – one that my within-sector comparison addresses via sector fixed effects. In addition, shift-share approaches (e.g. Autor, Dorn and Hanson 2013) are also subject to this concern, as Borusyak, Hull and Jaravel (2022) show.[2] Shift-shares leverage differences in local industrial composition: regions are more exposed when their baseline employment is concentrated in sectors hit by larger national shocks. My design instead generates variation in protection across states within the same sector through differences in baseline import exposure. Unlike shift-share designs, my approach allows me to absorb nationwide sector-level shocks with sector fixed effects.[3] For example, if policymakers targeted sectors that were declining because of nationwide technological trends, this would potentially confound both sector-level and shift-share analyses, but would not confound my design because sector fixed effects absorb nationwide sector-level trends.

For my design to be invalidated, policymakers would instead need to set nationwide sector-level tariffs in response to differential trends across states within the same sector. That is, they would need to target sectors not simply because the sector was declining nationally, but because high-import states within that sector were on different trajectories

substitute away from Chinese imports, some of this demand may shift to communications equipment producers in Oregon. I study these spillovers in Section 4.

[2]This is true for designs based on the "exogenous shocks" view of Borusyak, Hull and Jaravel (2022). The "exogenous shares" view of Goldsmith-Pinkham, Sorkin and Swift (2020) requires that industrial composition is uncorrelated with future trends. This is unlikely to hold in my context, as locations specializing in different sectors are likely to be exposed to different shocks such as immigrant flows, automation and unionization drives. See discussion in Borusyak, Hull and Jaravel (2025).

[3]Appendix B.2 provides an explicit example and formally compares this design with shift-share approaches.

than low-import states, after accounting for both state-wide and sector-wide shocks. This is a more specific form of targeting than targeting particular states or sectors, and is contrary to the stated justifications of the policy, as I go into more detail in Section 2.3. Consistent with this, I find no evidence of a differential pre-trend, and the policy is balanced on observable state-sector characteristics.

In the first stage, the tariff instrument strongly predicts import declines: state-sector pairs with greater pre-tariff import exposure experienced larger reductions in imports after the tariff increases, consistent with the gravity framework. In the second stage, I find that state-sector pairs with larger tariff-induced declines in imports experienced larger increases in output. In my baseline specification, a \$1 decline in imports leads to a \$0.63 increase in output. I also find positive effects on intermediate input use and capital expenditures, suggesting that the output response is not driven purely by market power.

These estimates should be interpreted as local, partial-equilibrium effects: they compare state-sector pairs with greater tariff exposure to state-sector pairs with less exposure. As emphasized by the micro-to-macro, or "missing intercept," literature (Muendler 2017; Chodorow-Reich 2019, 2020; Guren et al. 2021), positive relative effects can be consistent with zero aggregate effects, or even negative nationwide effects. For example, workers or firms may move from less-protected state-sector pairs toward more-protected ones, so the observed relative gains may represent a reallocation of economic activity rather than an increase in aggregate output. I therefore estimate a number of general-equilibrium channels and spatial spillovers to shed light on the implied aggregate effects arising from import protection.

First, I examine migration flows across state-sector pairs. I use a spatial migration model to derive a reduced-form estimating equation for how baseline import exposure interacted with national tariff shocks affected worker reallocation across locations and sectors. I find no statistically significant effect on migration flows, perhaps due to uncertainty surrounding the policy and its persistence. Next, I examine demand spillovers across space. Demand may shift from foreign suppliers toward domestic producers in other states, so the gains from protection may extend beyond the directly exposed state-sector pair. Using pre-existing spatial linkages from the Commodity Flow Survey, together with the gravity framework, I construct a measure of tariff-induced spillover demand across state-sector pairs. I find that this spillover measure has a positive effect on output, marginally significant at the 10 percent level, suggesting that some gains from protection diffuse across state borders. Finally, I examine aggregate channels that are not captured by the local estimates. Tariffs generated revenue for the federal government, which may have supported additional spending through fiscal multiplier effects. At the same time, tariffs raised import prices and may have reduced real incomes, dampening aggregate demand. I account for both channels when translating the local estimates into an implied nationwide effect. Putting these forces together, their combined contribution is positive, suggesting that the implied nationwide effect is larger than the local cross-sectional estimate. In my preferred aggregation, import protection increased manufacturing output by $74.9 billion.

However, these estimates isolate only the import-protection channel. Several trading partners retaliated against the 2018 tariffs, targeting roughly $127 billion of U.S. exports. Unlike import protection, where I find no evidence of a differential pretrend, export retaliation appears correlated with prior trends, suggesting selective targeting by trading partners, consistent with Fetzer and Schwarz (2021). I therefore model this pre-trend directly to isolate the effect of export retaliation net of differential pre-existing trends, following the approach in Rambachan and Roth (2023). My preferred estimate implies that a $1 decline in exports leads to a $3.40 decline in manufacturing output at the state-sector level, outweighing the import-protection effect. A back-of-the-envelope aggregation of my preferred estimate implies that export retaliation reduced manufacturing output by $178 billion, more than offsetting the $74.9 billion gain from import protection.

Roadmap. Section 2 discusses the policy context, data, and empirical framework.

Section 3 presents the main results on import protection. Section 4 estimates a number of general equilibrium channels and spatial spillovers to gauge implied aggregate effects. Section 5 studies export retaliation.

2 Empirical Framework

We are interested in the following specification relating local production and imports:

$$\Delta Y_{ik} = \beta \Delta M_{ik} + \theta G_{ik} + \Delta\delta_i + \Delta\delta_k + \Delta e_{ik} \quad (1)$$

M_{ik} is the value of sector k products imported by state i, and Y_{ik} is the production value of sector k in state i. Δ is the difference operator (i.e. $\Delta Y_{ik} = Y_{ik,2019} - Y_{ik,2017}$). $\Delta\delta_i$ and $\Delta\delta_k$ are state and sector fixed effects. β captures the output response to a \$1 increase in imports. Because the specification is in first differences, it absorbs time-invariant state-sector factors, such as persistently low productivity of sector k in state i.

G_{ik} is a vector of controls, most notably containing exposure to export retaliation, which I define in Section 2.3. Because the 2018 import tariffs triggered retaliatory tariffs on U.S. exports, state-sector pairs receiving import protection may also have been exposed to foreign demand shocks. I therefore control for retaliatory tariff exposure throughout the baseline specifications, so that the import coefficient isolates the import-protection channel holding retaliation exposure fixed. I study the retaliation channel directly in Section 5.

The remaining concern is that changes in imports may be correlated with changes in unobserved state-sector shocks. Running OLS on Equation 1 may therefore suffer from reverse causality or omitted variables bias. For example, steel producers in Pennsylvania may receive negative productivity shocks, leading to reductions in local steel production and increases in foreign steel imports to Pennsylvania.

To address this concern I turn to an instrument motivated by the 2018 policy episode.

I interact the level of sector imports by state in 2017 with the 2018 tariff change. Mathematically,

$$Z_{ik}^{M} = M_{ik,2017}\Delta\log(1+t_k)$$

where t_k is the nationwide tariff imposed on sector k. The basic idea is that states with lots of imports from sector k will be most affected by the nationwide tariff hike on sector k, since there is greater scope to substitute away from imported sector k goods.[4] Therefore, the tariff should provide relatively more protection to domestic producers in more import-exposed state-sector pairs. The next subsection formalizes this intuition via a first order approximation to a gravity equation.

2.1 Structural Motivation of IV

Consider the following gravity equation, which models the determinants of imports. The gravity equation can be microfounded by Anderson (1979), Krugman (1980), Eaton and Kortum (2002), and Melitz (2003):[5]

$$M_{ik} = (p_{ik}^{f})^{1-\sigma} P_{ik}^{\sigma-1} E_{ik} \tag{2}$$

M_{ik} is imports by state i in sector k, p_{ik}^{f} is the delivered price of the foreign variety, P_{ik} is the CES price index in state-sector market ik,[6] E_{ik} is total expenditure in that market, and σ is the elasticity of substitution.

Let preferences across sectors be Cobb-Douglas. Thus, $E_{ik} = \mu_{ik}E_i$ where E_i is total expenditure in state i and μ_{ik} is the Cobb-Douglas preference weight state i places on sector

[4] To see this in the extreme case, suppose a state imports \$0 worth of sector k products. In a gravity framework, a nationwide tariff increase in sector k is essentially redundant, as there are no imports to substitute away from towards domestic products. (The gravity framework abstracts from threat-of-entry effects, whereby tariffs may affect domestic producers even in markets with no current imports.)

[5] See Head and Mayer (2014) for discussion on these microfoundations and others.

[6] The CES price index is $P_{ik} = \left[\sum_{s\in\mathcal{S}_{ik}}(p_{ik}^{s})^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$, where $\mathcal{S}_{ik}$ is the set of available varieties.

k goods. Note this allows different states to have different preferences across sectors.

I follow the standard multiplicative/iceberg assumption, and define the delivered foreign price as:

$$p_{ik}^f = p_k^f \tau_{ik}^f (1 + t_k)$$

where p_k^f is the foreign producer price at the factory gate, τ_{ik}^f captures non-tariff trade costs, most notably transportation costs, and t_k is the ad valorem tariff rate.

Log-differentiating Equation (2):

$$d \log M_{ik} = (1 - \sigma) d \log p_{ik}^f + (\sigma - 1) d \log P_{ik} + d \log E_{ik}$$

Since the delivered foreign price is $p_{ik}^f = p_k^f \tau_{ik}^f (1 + t_k)$, holding the foreign producer price and non-tariff trade costs fixed implies:

$$d \log p_{ik}^f = d \log(1 + t_k)$$

In Appendix A.1, I show:

$$d \log P_{ik} = \lambda_{ik}^f d \log(1 + t_k)$$

Intuitively, the sector k price index rises more in states with larger import shares when tariffs increase, as a larger share of sector k goods are affected by the price increase induced by tariffs.[7] Finally, I assume preferences do not change in this period, so $d \log \mu_{ik} = 0$. This leaves changes in total expenditures in state i, $d \log E_i$, which will be absorbed by the state fixed effect $\Delta \delta_i$. Therefore, the instrument isolates a partial-equilibrium substitution effect: the predicted change in imports holding total state expenditure fixed.

[7]Suppose state i imports no foreign sector k goods, i.e. $\lambda_{ik}^f = 0$. Then a tariff increase has no direct effect on the local sector k price index as no goods are affected by the price increase from the tariff.

$$d \log E_{ik} = \underbrace{d \log \mu_{ik}}_{0} + \underbrace{d \log E_i}_{\Delta \delta_i}$$

Putting these together we obtain:

$$d \log M_{ik} = (1-\sigma)(1-\lambda^f_{ik}) d \log(1+t_k)$$

Finally, using $dM_{ik} = M_{ik} d \log M_{ik}$ and replacing the differential with a finite change using a first-order approximation around the 2017 baseline, the predicted level change in imports is:

$$\Delta M_{ik} \approx (1-\sigma)(1-\lambda^f_{ik,2017}) M_{ik,2017} \Delta \log(1+t_k) \tag{3}$$

This equation shows that, while the tariff shock is national within a sector, the predicted level change in imports varies across states because baseline imports $M_{ik,2017}$ differ across states. States that initially import more of sector k products in 2017 (i.e. high $M_{ik,2017}$) have more scope to substitute away from such imports when tariffs rise, with this effect larger in magnitude when the elasticity of substitution σ is higher. Equation (2) clarifies why baseline sector imports differ across states. First, states closer to major foreign suppliers have lower trade costs τ^f_{ik} and therefore higher import levels, as illustrated by Washington state's proximity to China relative to Massachusetts. States may also differ in how much they consume sector k goods, captured by E_{ik}, or in local market conditions that make foreign varieties more attractive, summarized by a high P_{ik}. These pre-existing differences in baseline imports are what allow the same national sector-level tariff shock to generate different predicted import changes across states.

The term $(1-\lambda^f_{ik,2017})$ is a price-index dampener.[8] The baseline first stage omits this term

[8] As tariffs raise p^f_{ik}, imports fall. But if foreign goods are a large part of the sector basket, the tariff also raises P_{ik}. That means the relative price p^f_{ik}/P_{ik} rises less than the foreign price alone, attenuating the import response (but never reversing it).

for simplicity; Section 3.2 presents results using the theoretically consistent instrument $(1 - \lambda^f_{ik,2017}) M_{ik,2017} \Delta \log(1 + t_k)$.

2.2 Identification

The IV orthogonality conditions needed for consistent estimation are:

$$\mathbb{E}(M_{ik,2017} \cdot \Delta \log(1 + t_k) \cdot \Delta e_{ik} \mid \Delta\delta_i, \Delta\delta_k, G_{ik}) = 0 \tag{4}$$

A sufficient condition for Equation 10 is (see Appendix B.1):

$$\mathbb{E}(\Delta \log(1 + t_k) \cdot \Delta e_{ik} \mid M_{ik,2017}, \Delta\delta_i, \Delta\delta_k, G_{ik}) = 0 \tag{5}$$

In words, nationwide tariff changes must be uncorrelated with *local* state-sector shocks to output, conditional on state and sector fixed effects and controls. Because the specification is in differences, targeting based on any persistent state-sector characteristics is not a violation. For example, tariffs may target sectors in states with historically high import exposures, low productivities or higher political salience.[9] This includes the accumulated legacy of earlier trade shocks: if a state-sector pair is small or unproductive in 2017 as a result of NAFTA or the China shock, targeting on that basis is differenced out.

Crucially, my design also allows tariffs to be targeted on the basis of nationwide sector trends—for instance, policymakers may protect sectors that are declining nationwide or because of a lobbying push by that sector—because the sector fixed effects $\Delta\delta_k$ absorb any such targeting. Similarly, the state fixed effects $\Delta\delta_i$ absorb any statewide shocks, addressing concerns that the administration may have targeted Democratic-leaning states. What would violate this condition is if tariffs were targeted at the idiosyncratic component of a sector's performance in particular states, conditional on statewide and sectorwide

[9] Political salience may reflect high unionization, legacy manufacturing employment, or industries tied to national-security narratives.

conditions. This would require a more specific form of targeting than targeting sectors or states alone. I assess this assumption using pretrend and balance tests later in this section.

It is useful to contrast Equation (11) with conditions required by other designs. A common approach in the trade literature is to compare sectors that received larger tariff changes to those that received smaller ones (e.g. Pavcnik (2002), Khandelwal and Topalova (2011), Flaaen and Pierce (2024)). The analogous condition for such cross-sector designs is $\mathbb{E}(\Delta \log(1+t_k) \cdot \Delta e_k \mid \Delta\delta) = 0$: nationwide tariff changes must be uncorrelated with *nationwide* sectoral shocks— rather than merely *local* sectoral shocks. Similarly, as Borusyak, Hull, and Jaravel (2022) show, shift-share designs that rely on exogenous shocks require conditions analogous to those of cross-sector comparisons, and are thus subject to the same concern.[10]

2.3 Policy Context

In November 2016, Donald Trump was elected President of the US, contrary to expectations. In 2018, tariffs were raised unilaterally by the executive branch using three different statutory authorities. Since all three are officially based on nationwide rather than state-specific concerns, any nationwide sectoral targeting is absorbed by the sector fixed effects $\Delta\delta_k$. However, I further examine each policy document in turn to assess whether state-specific considerations may have played a role.

Section 301 tariffs, imposed on imports from China, were justified by alleged unfair Chinese trade practices, including industrial policies, technology transfers, and intellectual property practices. China has been accused of engaging in such practices for at least a decade prior: the United States Trade Representative first elevated China to its Special 301 Priority Watch list in 2005, spanning the Bush and Obama administrations. Thus, the

[10] This is true for designs based on the "exogenous shocks" view of Borusyak, Hull and Jaravel (2022). The "exogenous shares" view of Goldsmith-Pinkham, Sorkin and Swift (2020) requires that industrial composition is uncorrelated with future trends. This is unlikely to hold in my context, as states specializing in different sectors are likely to be exposed to different shocks such as immigrant flows, automation, and unionization drives. See discussion in Borusyak, Hull and Jaravel (2025).

application of tariffs in 2018 as a countermeasure appears to be driven by a political shift in the executive branch rather than by contemporaneous domestic state-sector trends.

Section 232 tariffs – used for the first time since the founding of the WTO in 1995 – were imposed on steel and aluminum after a Secretary of Commerce report declared these imports a threat to national security. A wide variety of justifications were used to come to this conclusion, but one was based on the absolute decline of domestic steel and especially aluminum production in the years preceding 2018. If the decline of steel and aluminum were concentrated in certain states, this raises the possibility of state-sector trends being correlated with the treatment. I drop Section 232 tariffs as a robustness check.

Section 201 tariffs authorizes the President to impose temporary safeguards to protect domestic sectors affected by sudden surges of imports. These tariffs were imposed on washing machines and solar panels. If the surge in imports were due to negative trends in the associated sectors, and if these firms were disproportionately located in certain states, this could potentially be correlated with the treatment. Therefore, I also examine robustness of the results to the exclusion of Section 201 tariffs.

In response to these tariffs, six trading partners imposed retaliatory tariffs on $127 billion of US exports across over 8,000 products between April and September 2018 (Fajgelbaum et al. 2020, Table I). China's retaliation was the largest at $92.5 billion, with soybeans alone accounting for $12.4 billion of the exports covered by China's June 2018 retaliation list (Bown, Jung and Lu 2018). The European Union targeted $8.2 billion, choosing products for their symbolic value: bourbon whiskey, Harley-Davidson motorcycles, and peanut butter (NPR, June 22, 2018). Canada, Mexico, Russia, and Turkey imposed additional measures. Fetzer and Schwarz (2021) document that retaliatory tariffs disproportionately targeted Republican-leaning counties, consistent with strategic political targeting by foreign governments. I therefore control for retaliatory tariff exposure in all specifications, instrumented by initial export volumes interacted with retaliatory tariff changes,[11] and discuss the output effects of retaliation in Section 5.

2.4 Data

Production, capital expenditures, materials expenditure, and employment come from the 2017 Economic Census and the Annual Survey of Manufactures (ASM) for 2013, 2015 and 2019. State-level imports and exports by NAICS commodity come from U.S. Census Bureau USA Trade Online merchandise trade statistics. U.S. import tariff changes and retaliatory tariff changes are taken from Flaaen and Pierce (2024), who provide tariff exposure at the four-digit NAICS level.

2.5 Pretrends and Balance Tests

Earlier in this section, I presented an ex-ante narrative for why tariffs were not correlated with state-sector trends. Here, I present a number of ex-post tests to bolster this assumption.

I begin with a standard pre-trend test, regressing pre-treatment output growth on the instrument:

$$Y_{ik,2017} - Y_{ik,2015} = \pi Z^M_{ik} + \zeta G_{ik} + \Delta\delta_i + \Delta\delta_k + \Delta\eta_{ik}$$

If $\hat{\pi}$ is significantly different from zero, this would suggest that the instrument is correlated with pre-existing trends—whether due to deliberate targeting or coincidence—casting doubt on the exclusion restriction.

As shown in Column 1 of Table 1, the import-protection instrument is statistically insignificant in the 2015–2017 pre-period, suggesting state-sector pairs receiving greater protection were not on differential pre-existing output trends relative to less-exposed state-sector pairs. Relative to the dispersion of pre-period output changes, the estimated pretrend coefficient is economically small: a \$1 million increase in the import-protection

[11] Mathematically, $Z^X_{ik} = X_{ik,2017} \cdot \Delta \log(1 + t^*_k)$, where $X_{ik,2017}$ is state i's exports of sector k in 2017 and t^*_k is the trade-weighted retaliatory tariff imposed by foreign governments on sector k.

Table 1: Pre-Trend and Balance Tests: Import-Protection Instrument

	(1) $Y_{2017} - Y_{2015}$	(2) Share College	(3) Share Foreign	(4) Share Female	(5) Avg. Age
Z_{ik}^{M}	-0.047	0.005	0.006	0.001	-0.020
	(0.032)	(0.004)	(0.004)	(0.004)	(0.133)
Dep. Var. Mean	1.12×10^{7}	0.237	0.158	0.303	45.130
Dep. Var. SD	7.19×10^{8}	0.158	0.142	0.155	3.707
Observations	2,718	1,035	1,035	1,035	1,035
State FE	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes
Retaliation Controls	Yes	Yes	Yes	Yes	Yes

Notes: Column 1 reports a reduced-form pre-trend regression of the 2015–2017 change in manufacturing output on the import-protection tariff instrument. Columns 2–5 report balance regressions of pre-treatment workforce characteristics on the same instrument. All specifications control for retaliatory tariff exposure and include state and sector fixed effects. Standard errors are in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

instrument predicts a change equal to less than 0.01 percent of a standard deviation of the pre-period outcome.

I also conduct a balance test. While the design does not require the instrument to be uncorrelated with baseline levels, if state-sector pairs receiving more protection are balanced on pre-treatment characteristics, this provides further suggestive evidence that Equation (11) holds. I regress a variety of workforce demographic variables from the 2017 American Community Survey —share college-educated, share foreign-born, share female, and average age—on the instrument. If state-sector pairs with higher instrument values differ systematically on these characteristics, this would raise concern that tariff changes were targeted towards state-sector pairs with certain demographic characteristics, thereby suggesting a violation of Equation (11).

Columns 2–5 of Table 1 report the balance test results. All coefficients are small and statistically insignificant, consistent with the instrument being uncorrelated with pre-treatment workforce composition. The coefficients are also modest relative to the dependent-variable means and standard deviations: the college-educated and foreign-born share coefficients are about 3–4 percent of a standard deviation, while the female-share and age coefficients are essentially zero.

3 Results

3.1 Baseline Results

The first stage equation is:

$$\Delta M_{ik} = \gamma Z_{ik}^{M} + \zeta G_{ik} + \Delta\delta_i + \Delta\delta_k + v_{ik} \tag{6}$$

where the instrument is $Z_{ik}^{M} = M_{ik,2017} \cdot \Delta \log(1+t_k)$.

Column 2 of Table 2 reports the first stage. A \$1 increase in the import IV leads to a \$0.57 fall in imports at the state-sector level. Thus, the 2018 tariff episode led to a larger decline in imports in state-sectors with higher initial import volumes, as predicted by the gravity model. At the average level of state-sector imports—\$515 million—a 1% increase in one plus the import tariff yields a \$2.95 million fall in imports.[12] The Sanderson-Windmeijer first-stage F-statistic is 50.20, above the conventional rule-of-thumb threshold of 10 for weak instruments (Stock and Yogo 2005).

Columns 1 and 3 of Table 2 present the OLS and 2SLS estimates corresponding to Equation (1). Both estimates agree on sign, but the OLS coefficient is smaller in magnitude than the 2SLS: a \$1 increase in imports is associated with a \$0.12 decline in output in OLS, compared with a \$0.63 decline in the 2SLS specification. One potential explanation for the OLS estimate's upward bias toward zero is booming state-sectors increasing both domestic production and demand for foreign inputs within the same NAICS-4 sector. At the mean level of state-sector imports—\$515 million—a 1% increase in one plus the import tariff raises output by \$1.86 million.[13]

[12] Mathematically, $\Delta M_{ik} = \hat{\gamma} \times \bar{M}_{ik,2017} \times \Delta \log(1+t_k) = -0.572 \times 515 \times 0.01 = -2.95$ million.
[13] $\hat{\beta} \times \hat{\gamma} \times \bar{M}_{2017} \times 0.01 = (-0.633)(-0.572)(515)(0.01) = 1.86$ million.

Table 2: OLS, First Stage and 2SLS Results

	OLS ΔY_{ik}	First Stage ΔM_{ik}	Second Stage ΔY_{ik}
Z_{ik}^M		-0.572^{***}	
		(0.027)	
ΔM_{ik}	-0.119^{***}		-0.633^{***}
	(0.020)		(0.131)
Observations	2,678	2,678	2,678
First Stage F-stat (SW)		50.20	
State FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Retaliation Controls	Yes	Yes	Yes

Notes: Column 1 reports OLS estimates of the 2017–2019 change in manufacturing output on the change in imports. Column 2 reports the first-stage regression of the change in imports on the import-protection tariff instrument. Column 3 reports 2SLS estimates of the change in manufacturing output on the instrumented change in imports. All specifications control for retaliatory tariff exposure and include state and sector fixed effects. The SW F-statistic is the Sanderson-Windmeijer first-stage F-statistic for imports. Standard errors are in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

3.2 Robustness and Confounds

Input Protection. Trade protection not only reduces imports competing with domestic producers, but may also raise the cost of imported inputs, harming downstream producers. I construct a measure of input tariff exposure using BEA input-output tables: $\Delta I_{ik} = \sum_{k' \neq k} \zeta_{k' \to k} \Delta M_{ik'}$, where $\zeta_{k' \to k}$ is the cost share of input k' in producing good k, excluding own-sector to separate from the direct import competition channel. The instrument is $Z_{ik}^I = \sum_{k' \neq k} \zeta_{k' \to k} Z_{ik'}^M$.

Column 1 of Table 3 reports the results. The import coefficient barely moves when controlling for input import penetration. The input coefficient is positive, consistent with cheaper inputs raising downstream output, though imprecisely estimated.

Endogeneity of Tariffs. As discussed in Section 2.3, Section 232 and Section 201 tariffs may reflect state-sector trends, potentially violating the exclusion restriction. Columns 2–4 of Table 3 report results dropping Section 232 tariffs, Section 201 tariffs, and both. The

Table 3: Robustness: Input Costs, Endogenous Tariff Sections, and Baseline Exposure

	(1) + Input Channel	(2) Drop §232	(3) Drop §201	(4) Drop Both	(5) + Base Imports
ΔM_{ik}	-0.679^{***}	-0.644^{***}	-0.654^{***}	-0.664^{***}	-0.541^{***}
	(0.145)	(0.137)	(0.148)	(0.154)	(0.132)
ΔI_{ik}	0.578				
	(0.444)				
$M_{ik,2017}$					0.018^{*}
					(0.011)
Observations	2,599	2,618	2,626	2,566	2,678
State FE	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes
Retaliation Controls	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable is the 2017–2019 change in manufacturing output at the state-sector level. Column 1 controls for input-cost exposure. Columns 2–4 drop sectors affected by Section 232 tariffs, Section 201 tariffs, and both sets of tariffs, respectively. Column 5 controls for baseline 2017 imports. All specifications instrument import changes with the import-protection tariff instrument, control for retaliatory tariff exposure, and include state and sector fixed effects. Standard errors are in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

import coefficient is stable across all specifications, ranging from -0.644 to -0.664.

Baseline Exposure. As Borusyak and Hull (2023) note, designs that interact predetermined exposure with plausibly exogenous shocks can still be confounded if baseline exposure is itself correlated with outcome trends. For example, a state with an initially uncompetitive sector may have a high price index P_{ik} and high import levels $M_{ik,2017}$ while continuing to face adverse supply shocks during the outcome period. I therefore include the initial level of imports, $M_{ik,2017}$, as a control.[14] With this control, identification comes from variation in tariff changes among state-sector pairs with similar baseline import exposure. As shown in Column 5 of Table 3, the estimated import effect remains negative and statistically significant after controlling for baseline imports.

Spatial Spillovers. When a state imports less of a given sector's products, consumers

[14] The first-order approximation in Section 2.1 also avoids having to recenter the instrument using the mean of the full counterfactual distribution of sectoral tariff shocks following Borusyak and Hull (2023). In this setting, specifying such a counterfactual assignment process for tariff shocks would be difficult to justify credibly.

may substitute not only toward producers in their own state, but also toward producers in other states. This represents a violation of the stable-unit treatment value assumption (SUTVA),[15] potentially resulting in a biased estimate for $\hat{\beta}$. In Table 7, I include a control for spatial spillovers motivated by gravity theory, which I go into more detail in Section 4.1. The 2SLS coefficient is virtually identical to the baseline after including this spillover control.

Theoretically Consistent First Stage. The baseline first stage in Equation (6) omits the price-index dampening term $(1 - \lambda_{ik}^{f})$ from the theoretically derived first-order approximation in Equation (3). I therefore also estimate a specification that includes this term, using the theoretically consistent instrument $(1 - \lambda_{ik,2017}^{f})Z_{ik}^{M}$, where $\lambda_{ik,2017}^{f}$ is the foreign import share of state-sector ik. The adjusted first-stage equation is:

$$\Delta M_{ik} = \gamma \left[(1 - \lambda_{ik,2017}^{f})Z_{ik}^{M}\right] + \zeta G_{ik} + \Delta\delta_i + \Delta\delta_k + v_{ik}, \tag{7}$$

where $Z_{ik}^{M} = M_{ik,2017}\Delta \log(1 + t_k)$.

Column 1 of Table 4 reports the first stage, which has a structural interpretation as a trade elasticity as shown in Section 2.1. A \$1 increase in $(1 - \lambda_{ik,2017}^{f})Z_{ik}^{M}$ leads to a \$2.05 decrease in imports. Column 2 shows the 2SLS results: a \$1 decrease in imports leads to a \$0.49 increase in output, compared with the baseline estimate of \$0.63 in Table 2.

3.3 Comparison to Other Estimates

A number of papers have studied the 2018 tariff episode on a variety of outcomes. I compare my estimates to the results in these other papers.

Trade Elasticities. The first-stage coefficient of the theoretically-consistent regression in Table 4 has a structural interpretation as a trade elasticity (see Section 2.1), with an implied estimate of 2.05. Both Fajgelbaum et al. (2020) and Amiti et al. (2019) study the

[15]SUTVA requires the outcome of a unit to be not affected by the treatment status of another unit.

Table 4: Theory-Consistent IV Results

	(1) First Stage ΔM_{ik}	(2) 2SLS ΔY_{ik}
$(1-\lambda^{f}_{ik,2017})Z^{M}_{ik}$	-2.054^{***}	
	(0.186)	
ΔM_{ik}		-0.486^{***}
		(0.161)
Observations	2,643	2,643
First Stage F-stat (SW)	46.09	
State FE	Yes	Yes
Sector FE	Yes	Yes
Retaliation Controls	Yes	Yes

Notes: Column 1 reports the first-stage regression of the 2017–2019 change in imports on the import-protection instrument interacted with one minus the foreign import share, $(1-\lambda^{f}_{ik,2017})Z^{M}_{ik}$. Column 2 reports the corresponding 2SLS estimate of the effect of import changes on manufacturing output. All specifications control for retaliatory tariff exposure and include state and sector fixed effects. Standard errors are in parentheses.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

same 2018 tariff episode and estimate trade elasticities of 2.5 and 5.9 respectively.

Compared to the wider trade literature, my estimate is somewhat on the lower side. Standard benchmarks include 3.6 in Eaton and Kortum (2002), 4.1 in Simonovska and Waugh (2014), and the 5–10 range surveyed by Anderson and van Wincoop (2004). However, these conventional estimates typically do not distinguish between short and long-run responses. Boehm, Levchenko, and Pandalai-Nayar (2023) estimate horizon-specific trade elasticities of 0.26 at impact, and 0.76 at one year. My estimate of 2.05 over a two-year horizon lies between these short-run estimates and the larger long-run benchmarks.

Manufacturing Results. Flaaen and Pierce (2024) also estimate the effects of the 2018 import protection episode on industrial production using a nationwide cross-sector design. My implied reduced-form coefficient—obtained by multiplying the first- and second-stage estimates—can be compared to theirs. I find a positive and significant coefficient of 0.362,[16] indicating that import protection boosts output, whereas Flaaen and

[16] I yield this by multiplying the first-stage and second-stage coefficients together, i.e. $\hat{\beta} \times \hat{\gamma} = (-0.633) \times$

Pierce (2024) find a negative and insignificant coefficient of -0.485.

However, these coefficients are not strictly comparable. Flaaen and Pierce's coefficient captures the *nationwide* sector-level response of output to protection, while my estimate captures the *state* sector-level response. Two wedges can separate these objects. First, if protection reallocates workers across state-sector pairs, local gains may overstate the aggregate effect. Second, when sector imports fall in one state, some of the resulting demand may be redirected toward producers in other states in the same sector, so local estimates may miss gains accruing outside the treated state.

Later in Section 4, I quantify these two channels. I find that there is no significant migration response, while spatial spillovers are positive, implying the sector-level coefficient should be **larger** than the state-sector level coefficient. This analysis suggests that the difference between my findings and Flaaen and Pierce's is more likely due to differing identification assumptions and sources of variation, rather than differences in the target parameter.

3.4 Mechanisms

As the Annual Survey of Manufactures reports revenue rather than physical quantities, the positive effect in Column 2 of Table 2 could in principle reflect market power: firms raise prices without expanding real production. Note the state fixed effect absorbs any general price inflation common across sectors within a state.

Ideally, I would estimate Equation (1) with physical output or prices on the left-hand side. Unfortunately, the ASM does not report quantity or price data at the state-sector level, and the Bureau of Labor Statistics does not publish producer price indices at the state level. Instead, I examine whether import protection increases input usage—materials expenditure and capital expenditure—as a proxy for whether real production expanded.

Table 5 reports the results. A \$1 reduction in imports raises materials expenditure

$(-0.572) = 0.362$.

Table 5: 2SLS Results: Effect of Imports on Materials and Capital Expenditures

	(1) ΔR_{ik}	(2) ΔK_{ik}
ΔM_{ik}	-0.146^{**}	-0.040^{*}
	(0.067)	(0.024)
Observations	2,063	1,481
State FE	Yes	Yes
Sector FE	Yes	Yes
Retaliation Controls	Yes	Yes

Notes: The dependent variables are the 2017–2019 changes in material expenditures and capital expenditures at the state-sector level. Both outcomes are measured in dollars. Each column reports a 2SLS regression of the outcome on the change in imports, instrumented by the import-protection tariff instrument. All specifications control for retaliatory tariff exposure and include state and sector fixed effects. Standard errors are in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

by \$0.15 (significant at 5%) and capital expenditure by \$0.04 (significant at 10%). This suggests that firms respond to import protection by purchasing more inputs and investing in additional capacity. This is difficult to reconcile with a pure market power story, in which firms would raise prices without expanding production or input usage.

Table 6 reports the effect of import protection on employment. Noting that the import variable is rescaled to millions for readability, a \$1 million reduction in imports is associated with 0.12 less workers at the state-sector level. The effect is statistically insignificant, which is striking given that capital and materials expenditure both increase. Several explanations could account for this pattern. First, firms may have used the additional revenue to invest in labor-saving technologies. Second, uncertainty surrounding the permanence of the tariffs may have discouraged hiring, a less reversible commitment than purchasing inputs.[17] Third, the US labor market was historically tight during this period, with unemployment at 3.7%, making it difficult for firms to find additional workers even if they wished to expand.

[17]Capital may be partially reversible because equipment can sometimes be resold, repurposed, or used beyond the tariff episode.

Table 6: 2SLS Results: Effect of Imports on Employment

	ΔL_{ik}
ΔM_{ik} (millions)	0.121
	(0.076)
Observations	2,624
State FE	Yes
Sector FE	Yes
Retaliation Controls	Yes

Notes: The dependent variable is the 2017–2019 change in employment at the state-sector level. Employment is measured in workers. The endogenous regressor is the change in imports, measured in millions of dollars (rather than dollars as in previous tables) and instrumented by the import-protection tariff instrument. All specifications control for retaliatory tariff exposure and include state and sector fixed effects. Standard errors are in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

4 Spillover and General Equilibrium Channels

The coefficient estimated in Section 3 is a local, partial equilibrium parameter: it captures the effect of a \$1 reduction in imports on output at the state-sector level, holding statewide and sectorwide conditions fixed. However, policymakers may instead be interested in the aggregate, nationwide effect of the 2018 tariffs on manufacturing output. As the micro-to-macro or "missing intercept" literature emphasizes (Chodorow-Reich (2019), Guren et al. (2021)), these two quantities are not equal in general, for two reasons.

First, the state fixed effect $\Delta\delta_i$ absorbs statewide changes in real incomes, including both the wage gains from increased local demand and the cost-of-living increases from higher import prices. The estimate in Table 2 holds these channels fixed, and thus is a partial equilibrium estimate.

Second, the estimate is identified from cross-state variation within sectors: more protected state-sectors gain relative to less protected ones. But a positive local effect is consistent with a null or even negative aggregate effect if protected locations crowd out less protected ones by drawing away labor or capital. Conversely, consumers facing reduced imports in one state may substitute towards varieties produced in other states, in which case the local estimate understates the aggregate effect. Therefore, we need to account for these spillovers across state-sector pairs.

I therefore try to quantify some of these channels in this section. Section 4.1 estimates demand spillovers across state borders. Section 4.2 examines whether tariff protection induced labor migration across states. Section 4.3 quantifies the output effects of government spending financed by tariff revenues. Section 4.4 estimates the income effects of tariff-induced price increases. Finally, Section 4.5 brings these channels together to assess the overall effect on manufacturing output.

4.1 Spatial Spillovers

When tariffs reduce foreign imports in a given state-sector, consumers may substitute not only towards local producers but also towards producers in other states. For example, when tariffs rise on Chinese electronics, California consumers substitute towards Oregon computer producers, which is not captured in Section 3.

To capture these cross-state linkages, I define a spatial spillover variable:

$$S_{ik}^{M} = \sum_{j \neq i} \underbrace{\frac{D_{ijk,0}}{\sum_{\ell} D_{\ell jk,0}}}_{\omega_{ijk}} \Delta M_{jk}$$

where $D_{ijk,0}$ is bilateral domestic shipments from state i to state j in sector k in 2017, $\sum_{\ell} D_{\ell jk,0}$ is total domestic shipments into destination state j from all US origins, and ΔM_{jk} is the change in foreign imports into state j in sector k. The weight $\omega_{ijk} = D_{ijk,0} / \sum_{\ell} D_{\ell jk,0}$ is thus state i's initial share of domestic sales into destination market j, k. This weight is higher if state i sells more of sector k to state j. Bilateral shipment shares are constructed from the 2017 Commodity Flow Survey.

This spillover term can be motivated by a gravity equation (see Appendix A.2). The key insight is that state i receives a larger spillover if the states it sells to experience a larger fall in imports. A negative coefficient indicates that when imports fall elsewhere, state i's output rises—consistent with cross-state demand substitution. Following the logic of Section 2, I construct an instrument for S_{ik}^M as $Z_{ik}^{spill} = \sum_{j \neq i} \omega_{ijk} Z_{jk}^M$.

Table 7: 2SLS Results: Effect of Imports and Spillovers on Output

	ΔY_{ik}
ΔM_{ik}	-0.637^{***}
	(0.116)
S_{ik}^M	-0.523^{*}
	(0.293)
Observations	2,676
State FE	Yes
Sector FE	Yes
Export Retaliation Controls	Yes

Notes: The dependent variable is the 2017–2019 change in manufacturing output at the state-sector level. Output is measured in dollars. The endogenous regressor is the change in imports, instrumented by the import-protection tariff instrument. S_{ik}^M captures tariff-induced import-protection spillovers from other states defined in Section 4.1. All specifications control for export retaliation and include state and sector fixed effects. Standard errors are in parentheses.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 7 reports the results. The direct import coefficient is essentially unchanged from the baseline (-0.637 vs. -0.633), indicating that the spillover channel does not confound the main estimate. The spillover coefficient is negative and significant at the 10% level: a \$1 reduction in imports in states that i sells to raises state i's output by \$0.52. This is consistent with cross-state demand substitution—when foreign goods become more expensive in destination markets, buyers substitute towards domestic suppliers in other states. The magnitude implies that the local estimates in Table 2 understate the aggregate effect of import protection, as they miss these positive cross-state demand spillovers.

4.2 Migration

If tariff protection attracts workers from other states, the local output gains documented in Table 2 partly reflect labor inflows from other states. Under constant returns to scale, such migration is zero-sum nationally: every worker gained by a protected state-sector is lost by an unprotected one, reducing the aggregate effect relative to the local estimate. This would thus serve to reduce the nationwide effect relative to the local effect.

I derive a bilateral migration equation from a discrete choice model of worker location (see Appendix A.3):

$$\Delta \log(f_{odk'}) = \alpha Z^M_{dk'} + \Delta\delta_{ok'} + \Delta\delta_{od} + \varepsilon_{odk'} \tag{8}$$

where $f_{odk'}$ is the flow of workers from origin state o to destination state d in sector k', $Z^M_{dk'}$ is the import instrument defined in Section 2, $\delta_{ok'}$ absorbs origin-by-sector effects, and δ_{od} absorbs origin-by-destination migration costs. Worker flow data $f_{odk'}$ is taken from the American Community Survey.

However, taking logs of the left-hand side drops zeros, which account for roughly 35% of bilateral flow cells. As a transparent way to handle these zeros without dropping them, I follow Chen and Roth (2024), who advocate for explicitly calibrating how much one values the extensive margin relative to the intensive margin.[18]

$$m(y) = \begin{cases} \log(y) & \text{if } y > 0 \\ -x & \text{if } y = 0 \end{cases}$$

I report results for $x \in \{0.1, 0.5, 1.0\}$. At $x = 0.1$, a new corridor appearing ($0 \to 1$ person) is valued at 0.1 log points—equivalent to a 10% increase in an existing flow. At

[18] Other approaches such as $\log(1 + f_{odk'})$ or $\text{arcsinh}(f_{odk'})$ transformations implicitly place a weight on the extensive margin relative to the intensive margin that depends on the units of measurement (Chen and Roth 2024). Poisson pseudo-maximum likelihood (Santos Silva and Tenreyro, 2006) offers another approach to handling zeros. However, it is inappropriate in my context because the dependent variable is a first difference of flows, which can be negative, and PPML requires non-negative outcomes.

$x = 0.5$ and $x = 1.0$, the same event is valued at 0.5 and 1.0 log points, equivalent to a 65% and 170% increase respectively. At the median flow of 144 workers, these correspond to increases of 15, 93, and 248 additional workers. Since a change from 0 to 1 worker is economically small, my preferred specification is $x = 0.1$.

Table 8: Migration Flow Regressions

	(1)	(2)	(3)
	$x = 0.1$	$x = 0.5$	$x = 1$
$Z^M_{dk'}$	-0.576	-0.621	-0.677
	(0.399)	(0.432)	(0.473)
Observations	2,971	2,971	2,971
Origin × Sector FE	Yes	Yes	Yes
Origin × Dest FE	Yes	Yes	Yes
Retaliation Controls	Yes	Yes	Yes

Notes: The dependent variable is the change in transformed bilateral migration flows, $\Delta m(f_{odk'})$, where $m(y) = \log(y)$ for positive flows and $m(0) = -x$ for zero flows. Columns report results for different values of the extensive-margin calibration x. The main regressor is the destination state-sector import-protection tariff instrument. All specifications control for retaliatory tariff exposure and include origin-by-sector and origin-by-destination fixed effects. Standard errors are clustered by origin-by-sector and reported in parentheses. Coefficients are reported in units of 10^{-10} for readability.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 8 reports the results. Across all extensive-margin calibrations, the migration response to tariff-induced import changes is economically small and statistically insignificant. At the preferred calibration ($x = 0.1$), the estimate implies that a $1 billion reduction in imports increases bilateral inflows by fewer than 0.06 workers. Given the two-year horizon of this study and the uncertainty surrounding the permanence of the tariffs, it is plausible that the policy was too short-lived and uncertain to induce cross-state relocation.

4.3 Fiscal Effects

Tariffs can generate government revenue, which, when spent, stimulates output through standard fiscal multiplier channels. The semi-elasticity of tariff revenue with respect to

$\log(1+t)$ is (see Appendix A.4 for derivation):

$$\frac{\partial R}{\partial \log(1+t)} = pQ\left[1 + t\rho(\epsilon_t^D + 1)\right]$$

where pQ is the initial import base, ρ is the passthrough of the tariff to the duty-inclusive price, ϵ_t^D is the price elasticity of import demand, and t is the tariff rate. Fajgelbaum et al. (2020) estimate $\epsilon_t^D = -2.53$ and find that passthrough is complete ($\rho = 1$), meaning foreign exporters do not adjust their pre-tariff prices in response to US tariffs.[19] Using the trade-weighted average tariff on targeted goods prior to the 2018 tariffs of $t = 0.026$ (Fajgelbaum et al. 2020, Table I) and the 2017 import base of targeted goods of $pQ = \$302{,}970$ million:

$$\Delta R = pQ[1 + t\rho(\epsilon_t^D + 1)] \times \Delta \log(1+t) = 302{,}970 \times 0.960 \times 0.01 = \$2{,}910\text{m}$$

A 1% increase in one plus the tariff thus yields \$2.91 billion in tariff revenues. Applying this first-order approximation to the full tariff change of $\Delta \log(1+t) = \log(1.166) - \log(1.026) = 0.128$ yields \$37.3 billion in tariff revenues.[20]

To translate tariff revenue into output effects, I apply fiscal multiplier estimates from the literature. Ramey (2011) reports deficit-financed spending multipliers ranging from 0.8 to 1.5, while Chodorow-Reich (2019) argues for 1.7.

$$\Delta Y^{fiscal} = \underbrace{\text{Spending Multiplier}}_{0.8-1.7} \times \Delta R = (0.8 - 1.7) \times 2{,}910 \approx \$2{,}328\text{m to } \$4{,}947\text{m}$$

A 1% increase in one plus tariffs generates \$2.3 billion to \$4.9 billion in output through the fiscal channel. Applying the first-order approximation to the full tariff change $\Delta \log(1+t) = \log(1.166) - \log(1.026) = 0.128$ yields a fiscal output effect ranging from \$29.8 billion

[19] The revenue-maximizing rate is 65.4% under these parameter values.

[20] This is roughly consistent with the \$34.3 billion computed by the full quantitative model in Fajgelbaum et al. (2020, Table VIII).

to \$63.3 billion. However, tariff revenue is not deficit-financed—it is financed by higher prices paid by consumers and firms. The spending multipliers cited above therefore overstate the net fiscal stimulus, as they do not account for the contractionary effect of the implicit tax. I address this in the next section by separately estimating the income effects of tariff-induced price increases.

4.4 Income Effects

Tariffs raise the price of imported goods[21], reducing consumers' real incomes and in turn their spending and ultimately output. In my baseline specification, this channel is absorbed by the state fixed effect $\Delta\delta_i$. To quantify its magnitude, I need two inputs: the passthrough of tariffs to consumer prices, and the marginal propensity to consume.

MPC estimates have been the subject of a large literature in macroeconomics. Parker et al. (2013) report widely-cited estimates ranging from 0.5 to 0.9 out of the 2008 stimulus checks. However, Orchard, Ramey and Wieland (2025) argue that after correcting for a number of biases, the micro MPC is approximately 0.3.

For the passthrough of tariffs to consumer prices, I would ideally run my baseline specification with a state-sector price index as the dependent variable. Unfortunately, the Bureau of Labor Statistics does not collect consumer price indices at the state level. I therefore rely on Amiti, Redding and Weinstein (2019, Table 4, Panel B), who estimate that the 2018 tariffs raised manufacturing producer prices by 1.030% in aggregate, accounting for both the output tariff and input tariff channels.[22] Since manufactured goods constitute approximately 19.5% of the consumer expenditure basket (Bureau of Labor Statistics, CPI Relative Importance Weights, December 2018), the implied effect on the overall consumer price level is $0.0805\% \times 0.195 = 0.0157\%$ per 1% increase in $(1 + t)$.

[21]Unless foreign exporters fully absorb the tariff by cutting their pre-tariff prices by the full amount of the tariff, an empirically very rare phenomenon and contrary to what Fajgelbaum et al. (2020) find in this episode.

[22]This is the effect of the full tariff change of $\Delta \log(1 + t) = 0.128$, implying a $1.030/12.8 = 0.0805\%$ increase in manufacturing PPI per 1% increase in $(1 + t)$.

Combining:

$$\Delta Y^{income} = -\underbrace{\text{MPC}}_{0.3-0.9} \times \underbrace{0.000157}_{\text{price effect}} \times \underbrace{Y}_{\$19.5\text{t}} \approx -\$0.9\text{b to } -\$2.8\text{b}$$

Thus, a 1% increase in one plus the tariff reduces output by $0.9 billion to $2.8 billion through the income channel. Applying the first-order approximation to the full tariff change of $\Delta \log(1+t) = 0.128$ yields income effects ranging from $-\$11.7$ billion (MPC $= 0.3$) to $-\$35.2$ billion (MPC $= 0.9$).

4.5 Discussion and Total Effect

I now combine the channels estimated in the preceding sections to compute the aggregate effect of import protection.

The fiscal channel generates $2.3 to $4.9 billion per 1% increase in $(1+t)$, while the income channel reduces output by $0.9 to $2.8 billion. Two features of the US economy explain why the fiscal channel tends to dominate. First, the US starts from a very low tariff baseline, generating large scope for revenue increases.[23] Combined with fiscal multipliers above unity, this translates into substantial output effects. Second, manufactured goods constitute a relatively small share of consumer expenditure in a services-heavy economy like the US, dampening the adverse real income effects.

Aggregating the direct import protection effect across all state-sector pairs, adding the fiscal channel, and subtracting the income channel:

$$\underbrace{\Delta Y}_{\text{Total}} = \underbrace{\hat{\beta} \times \hat{\gamma} \times \sum_{i,k} Z_{ik}^M}_{\text{Direct trade effect}} + \underbrace{\Delta Y^{fiscal}}_{\text{Fiscal channel}} - \underbrace{\Delta Y^{income}}_{\text{Income channel}}$$

The total import instrument sums to $\sum_{i,k} Z_{ik}^M = \$206.8$ billion. Multiplying by the first-

[23] According to Fajgelbaum et al. (2020), the baseline tariff rate was 2.6%, substantially below the revenue-maximizing rate of 65.4% under the parameters in Section 4.3.

stage coefficient yields the tariff-induced import decline: $\hat{\gamma} \times \sum_{i,k} Z_{ik}^{M} = -0.572 \times 206.8 = -\$118.3$ billion. Applying the second-stage coefficient, the implied output gain from import protection is $\hat{\beta} \times (-118.3) = -0.633 \times (-118.3) = +\$74.9$ billion. Adding the fiscal channel ($+\$29.8$ to $+\$63.3$ billion) and subtracting the income channel ($-\$11.7$ to $-\$35.2$ billion) yields a total import protection effect ranging from $+\$69.5$ to $+\$127.5$ billion.

The remaining channels reinforce these conclusions. Migration flows are economically negligible and statistically insignificant (Table 8), suggesting minimal crowding out of less protected state-sectors over this horizon. The spatial spillover estimates, while statistically insignificant at the 5% level, are consistent with cross-state demand substitution that would further increase the aggregate effect (Table 7). The estimates above are therefore conservative in that they omit these positive spillovers. Taken together, the channels estimated above have a positive net effect, suggesting that the implied aggregate effect of import protection exceeds the local partial-equilibrium estimate.

5 Export Retaliation

The preceding sections found that import protection boosted manufacturing output locally and suggested positive aggregate effects for the import-protection channel. However, these estimates held fixed the export-retaliation channel by controlling for exposure to retaliatory tariffs to isolate the import protection channel. As discussed in Section 2.3, US tariffs triggered retaliatory responses from six trading partners, collectively targeting $127 billion of US exports. To provide a fuller appraisal of the 2018 policy episode, I now estimate the output cost of this retaliation using the export channel of the same 2SLS specification presented in Section 3.

5.1 Pre-Trends in Export Retaliation

For convenience, I reproduce the definition of the export retaliation variable here (originally defined in Section 2.3), which will be used to instrument for exports:

$$Z_{ik}^X = X_{ik,2017} \Delta \log(1 + t_k^*)$$

where $X_{ik,2017}$ is state i's exports of sector k in 2017 and t_k^* is the trade-weighted retaliatory tariff imposed by foreign governments on sector k.

I first present a pre-trend and balance test on the export-retaliation instrument, parallel to the tests for the import protection instrument in Section 2.5. Reassuringly, Columns 2–5 of Table 9 show no significant correlation between the export-retaliation instrument and pre-treatment workforce characteristics, suggesting that these demographic characteristics were not a key target of retaliating countries. However, unlike the import-protection instrument, which shows no pre-trend (Table 1), the export instrument exhibits a significant positive pre-trend in the 2015–2017 window ($\hat{\pi}_2 = 1.340$, $p < 0.01$): state-sector pairs later subject to higher retaliatory tariffs were already growing faster before the trade war (Column 1 of Table 9). This is consistent with strategic targeting by foreign governments, which Fetzer and Schwarz (2021) have documented. The immediate pre-trend suggests foreign trading partners may have selected fast-growing US export sectors to maximize political pressure.

As the following section shows, export retaliation reduces output through the export channel. Because state-sector pairs more exposed to retaliation were already growing faster before the trade war, adjusting for a continuation of this pre-trend would make the estimated retaliation effect even more negative. I nevertheless conduct a formal sensitivity analysis using the methodology of Rambachan and Roth (2023), which also allows for mean reversion: the post-period violation of parallel trends may have the opposite sign of the pre-period violation instead of the same sign.

Table 9: Pre-Trend and Balance Tests: Export-Retaliation Instrument

	(1) $Y_{2017} - Y_{2015}$	(2) Share College	(3) Share Foreign	(4) Share Female	(5) Avg. Age
Z_{ik}^X	1.340^{***}	0.004	0.002	0.003	0.012
	(0.331)	(0.003)	(0.003)	(0.003)	(0.106)
Dep. Var. Mean	1.12×10^7	0.237	0.158	0.303	45.130
Dep. Var. SD	7.19×10^8	0.158	0.142	0.155	3.707
Observations	2,718	1,035	1,035	1,035	1,035
State FE	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes
Import Protection Control	Yes	Yes	Yes	Yes	Yes

Notes: Column 1 reports a reduced-form pre-trend regression of the 2015–2017 change in manufacturing output on the export-retaliation instrument. Columns 2–5 report balance regressions of pre-treatment workforce characteristics on the same instrument. All specifications control for import-protection exposure and include state and sector fixed effects. Standard errors are in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

5.2 Estimates

I first present the baseline estimate without the pre-trend correction. I estimate the following specification:

$$\Delta Y_{ik} = \theta \Delta X_{ik} + \zeta G_{ik} + \Delta \delta_i + \Delta \delta_k + \Delta e_{ik} \tag{9}$$

where Y_{ik} is output and X_{ik} is exports of sector k in state i. ΔX_{ik} is instrumented by $Z_{ik}^X = X_{ik,2017} \cdot \Delta \log(1 + t_k^*)$, the interaction of initial export volumes with the trade-weighted retaliatory tariff change in sector k. G_{ik} is a vector of controls including the import instrument, $\Delta\delta_i$ and $\Delta\delta_k$ are state and sector fixed effects, and θ captures the output response to a \$1 change in exports.

Column 2 of Table 10 reports the first stage. A \$1 increase in the export-retaliation instrument leads to a \$1.61 fall in exports at the state-sector level. Thus, the retaliatory tariff episode led to larger export declines in state-sectors with higher initial export volumes in sectors targeted by retaliation.

Columns 1 and 3 of Table 10 present the OLS and 2SLS estimates. Both estimates agree on sign, but the OLS coefficient is smaller in magnitude than the 2SLS: a \$1 increase in exports is associated with a \$1.45 increase in output in OLS, compared with a \$3.40 increase in the 2SLS specification. Of the \$3.40 output response, \$1 reflects the mechanical loss of export revenue, while the remaining \$2.40 is consistent with agglomeration, learning-by-exporting, and local multiplier effects.[24] For comparison, Section 3 estimates a per-dollar output effect of import protection of \$0.633, roughly one-fifth the magnitude of the per-dollar output loss implied by export retaliation.

Table 10: OLS, First Stage and 2SLS Results: Export Retaliation

	OLS ΔY_{ik}	First Stage ΔX_{ik}	Second Stage ΔY_{ik}
Z_{ik}^X		-1.606^{***}	
		(0.195)	
ΔX_{ik}	1.445^{***}		3.402^{***}
	(0.030)		(0.405)
Observations	2,678	2,678	2,678
First Stage F-stat (SW)		36.75	
State FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Import Protection Control	Yes	Yes	Yes

Notes: Column 1 reports OLS estimates of the 2017–2019 change in manufacturing output on the change in exports. Column 2 reports the first-stage regression of the change in exports on the export-retaliation instrument, $Z_{ik}^X = X_{ik,2017}\Delta\log(1+t_k^*)$. Column 3 reports 2SLS estimates of the change in manufacturing output on the instrumented change in exports. All specifications control for import-protection exposure and include state and sector fixed effects. The SW F-statistic is the Sanderson-Windmeijer conditional first-stage F-statistic for exports. Standard errors are in parentheses.

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Now, to account for the pre-trend, I turn to the approach of Rambachan and Roth (2023). Let γ^{post} capture, in the absence of retaliation, how state-sector pairs more exposed to export retaliation would have grown relative to less-exposed state-sector pairs in 2017–2019. Let γ^{pre} be the analogous difference in growth in the pre-period, measured using

[24]Moretti (2010) estimates that each tradable job generates 1.6 additional local jobs through demand spillovers, implying a total employment multiplier of 2.6. De Loecker (2013) documents productivity gains of 7–8% from entering export markets, suggesting that losing export access may reduce firm productivity.

2015–2017 output changes. The idea is to use the observed pre-trend γ^{pre} to bound the unobserved γ^{post}. Formally:

$$|\gamma^{post}| \leq M|\gamma^{pre}|$$

where M is a sensitivity parameter governing the size of post-period violations. For example, $M = 0$ imposes no post-period violation and therefore recovers the unadjusted treatment effect, $M = 1$ allows the post-period violation to be as large as the observed pre-period violation, and $M = 2$ allows it to be twice as large. Because the restriction is imposed in absolute value, the post-period violation may have the opposite sign of the pre-trend, allowing for mean reversion.

Details on the implementation of the Rambachan and Roth approach in this setting can be found in Appendix B.3. I present the results of the analysis in Table 11. As we see below, from $M = 0$ up to $M = 1.5$, the confidence interval for the export effect does not overlap with the confidence interval for the import effect in absolute value: even the lower bound of the export effect exceeds the upper bound of the import effect. For larger values of M, the intervals overlap, so the data no longer rule out equality between the two effects. However, even at $M = 2$ and $M = 2.5$, the estimates remain centered on substantially larger export-retaliation losses than import-protection gains.

Therefore, the per-dollar output loss from export retaliation exceeds the per-dollar output gain from import protection unless the post-period violation reverses the sign of the pre-trend and is more than 1.5 times as large. A post-period trend of the same sign would instead amplify the export-retaliation effect relative to the import-protection effect, so only a sign reversal can close the gap. Since $M = 1$ allows a full mean reversion in levels — the cumulative deviation of exposed state-sector pairs from the parallel-trends path returns to zero — $M = 1.5$ requires that exposed state-sector pairs not only revert to the parallel-trends path but overshoot it by 50 percent of the original deviation. The conclusion is therefore robust to post-period violations up to 1.5 times the observed pre-

Table 11: Sensitivity to Export-Retaliation Pre-Trends

	Import Effect	Export Effect
M	$\hat{\beta}$	$\hat{\theta}$
0	$[-0.731, -0.536]$	$[2.848, 3.956]$
1	$[-0.920, -0.347]$	$[1.767, 5.038]$
1.5	$[-1.040, -0.226]$	$[1.082, 5.723]$
2	$[-1.161, -0.106]$	$[0.394, 6.410]$
2.5	$[-1.285, 0.018]$	$[-0.312, 7.117]$

Notes: The table reports Rambachan and Roth (2023) sensitivity intervals using the 2015–2017 export-retaliation pre-trend. The intervals are obtained by applying the sensitivity procedure to the reduced-form export-retaliation coefficient and then transforming the resulting confidence sets into implied structural effects using the estimated first-stage matrix. The import effect $\hat{\beta}$ is the effect of imports on output, while the export effect $\hat{\theta}$ is the effect of exports on output.

trend — strictly beyond full mean reversion in levels.

To put these estimates in nationwide terms, I use a back-of-the-envelope calculation. The total export instrument sums to $\sum_{i,k} Z_{ik}^X = \$32.5$ billion. Multiplying by the first-stage coefficient yields a tariff-induced export decline of $-1.606 \times 32.5 = -\$52.3$ billion. At the unadjusted estimate of $\hat{\theta} = 3.40$, the aggregate output loss from export retaliation is $3.40 \times 52.3 = \$177.8$ billion, exceeding the aggregate import-protection gain of $74.9 billion from Section 4 by a factor of roughly 2.4. Even at the lower bound of the $M = 1$ confidence interval—corresponding to full mean reversion in levels—the implied loss is $92.4 billion, still exceeding the import-protection gain.

6 Conclusion

References

Amiti, Mary, Stephen J. Redding, and David E. Weinstein, "The Impact of the 2018 Tariffs on Prices and Welfare," *Journal of Economic Perspectives*, 2019, *33* (4), 187–210.

Anderson, James E., "A Theoretical Foundation for the Gravity Equation," *American Economic Review*, March 1979, *69* (1), 106–116.

_ and Eric van Wincoop, "Trade Costs," *Journal of Economic Literature*, September 2004, *42* (3), 691–751.

Autor, David H., David Dorn, and Gordon H. Hanson, "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," *American Economic Review*, 2013, *103* (6), 2121–2168.

Boehm, Christoph E., Andrei A. Levchenko, and Nitya Pandalai-Nayar, "The Long and Short (Run) of Trade Elasticities," *American Economic Review*, April 2023, *113* (4), 861–905.

Borusyak, Kirill and Peter Hull, "Non-random Exposure to Exogenous Shocks: Theory and Applications," *Econometrica*, November 2023, *91* (6), 2301–2338.

_ , _ , and Xavier Jaravel, "Quasi-Experimental Shift-Share Research Designs," *The Review of Economic Studies*, 2022, *89* (1), 181–213.

_ , _ , and _ , "A Practical Guide to Shift-Share Instruments," *Journal of Economic Perspectives*, 2025, *39* (1), 181–204.

Bown, Chad P., Euijin Jung, and Zhiyao Lu, "China's Retaliation to Trump's Tariffs," Peterson Institute for International Economics, Trade and Investment Policy Watch June 2018.

Bureau of Economic Analysis, "Input-Output Accounts Data: Use Tables," U.S. Department of Commerce, Washington, DC 2017.

Chen, Jiafeng and Jonathan Roth, "Logs with Zeros? Some Problems and Solutions," *The Quarterly Journal of Economics*, May 2024, *139* (2), 891–936.

Chodorow-Reich, Gabriel, "Geographic Cross-Sectional Fiscal Spending Multipliers: What Have We Learned?," *American Economic Journal: Economic Policy*, May 2019, *11* (2), 1–34.

—, "Regional Data in Macroeconomics: Some Advice for Practitioners," *Journal of Economic Dynamics and Control*, 2020, *115.*

De Loecker, Jan, "Detecting Learning by Exporting," *American Economic Journal: Microeconomics*, August 2013, *5* (3), 1–21.

Eaton, Jonathan and Samuel Kortum, "Technology, Geography, and Trade," *Econometrica*, September 2002, *70* (5), 1741–1779.

Executive Office of the President, "Proclamation 9693 of January 23, 2018: To Facilitate Positive Adjustment to Competition from Imports of Certain Crystalline Silicon Photovoltaic Cells," 83 Fed. Reg. 3541 (January 25, 2018) 2018.

—, "Proclamation 9694 of January 23, 2018: To Facilitate Positive Adjustment to Competition from Imports of Large Residential Washers," 83 Fed. Reg. 3553 (January 25, 2018) 2018.

Fajgelbaum, Pablo D., Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal, "The Return to Protectionism," *The Quarterly Journal of Economics*, 2020, *135* (1), 1–55.

Fetzer, Thiemo and Carlo Schwarz, "Tariffs and Politics: Evidence from Trump's Trade Wars," *The Economic Journal*, May 2021, *131* (636), 1717–1741.

Flaaen, Aaron and Justin Pierce, "Disentangling the Effects of the 2018–2019 Tariffs on a Globally Connected U.S. Manufacturing Sector," *The Review of Economics and Statistics*, September 2024, pp. 1–45. Advance online publication.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift, "Bartik Instruments: What, When, Why, and How," *American Economic Review*, 2020, *110* (8), 2586–2624.

Guren, Adam M., Alisdair McKay, Emi Nakamura, and Jón Steinsson, "What Do We Learn from Cross-Regional Empirical Estimates in Macroeconomics?," *NBER Macroeconomics Annual*, 2021, *35*, 175–223.

Head, Keith and Thierry Mayer, "Gravity Equations: Workhorse, Toolkit, and Cookbook," in Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, eds., *Handbook of International Economics*, Vol. 4, Amsterdam: Elsevier, 2014, chapter 3, pp. 131–195.

Krugman, Paul, "Scale Economies, Product Differentiation, and the Pattern of Trade," *American Economic Review*, December 1980, *70* (5), 950–959.

McFadden, Daniel, "Modelling the Choice of Residential Location," in Anders Karlqvist, Lars Lundqvist, Folke Snickars, and Jörgen W. Weibull, eds., *Spatial Interaction Theory and Planning Models*, Amsterdam: North-Holland, 1978, pp. 75–96.

Melitz, Marc J., "The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity," *Econometrica*, 2003, *71* (6), 1695–1725.

Moretti, Enrico, "Local Multipliers," *American Economic Review*, May 2010, *100* (2), 373–377.

Muendler, Marc-Andreas, "Trade, Technology, and Prosperity: An Account of Evidence from a Labor-Market Perspective," WTO Staff Working Papers ERSD-2017-15, World Trade Organization (WTO), Economic Research and Statistics Division 2017.

NPR, "EU Tariffs Take Effect, Retaliating for Trump's Tariffs on Steel and Aluminum," NPR June 2018. June 22, 2018.

Office of the United States Trade Representative, "2005 Special 301 Report," Report, Office of the United States Trade Representative, Washington, DC 2005.

_ , "Findings of the Investigation into China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property, and Innovation under Section 301 of the Trade Act of 1974," Report, Office of the United States Trade Representative, Washington, DC March 2018. Released March 22, 2018.

Orchard, Jacob, Valerie A. Ramey, and Johannes F. Wieland, "Micro MPCs and Macro Counterfactuals: The Case of the 2008 Rebates," *The Quarterly Journal of Economics*, 2025, *140* (3), 2001–2052.

Parker, Jonathan A., Nicholas S. Souleles, David S. Johnson, and Robert McClelland, "Consumer Spending and the Economic Stimulus Payments of 2008," *American Economic Review*, October 2013, *103* (6), 2530–2553.

Pavcnik, Nina, "Trade Liberalization, Exit, and Productivity Improvements: Evidence from Chilean Plants," *The Review of Economic Studies*, January 2002, *69* (1), 245–276.

PBS NewsHour, "Trump Has Touted Gilded Age Tariffs, an Era Which Saw Industrial Growth Together with Poverty," PBS NewsHour 2025.

Rambachan, Ashesh and Jonathan Roth, "A More Credible Approach to Parallel Trends," *The Review of Economic Studies*, September 2023, *90* (5), 2555–2591.

Ramey, Valerie A., "Can Government Purchases Stimulate the Economy?," *Journal of Economic Literature*, 2011, *49* (3), 673–685.

Ruggles, Steven, Sarah Flood, Matthew Sobek, Daniel Backman, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, Renae Rogers, Jonathan Schroeder, and

Kari C.W. Williams, "IPUMS USA: Version 16.0 [dataset]," Minneapolis, MN: IPUMS 2025. American Community Survey microdata.

Sanderson, Eleanor and Frank Windmeijer, "A Weak Instrument F-Test in Linear IV Models with Multiple Endogenous Variables," *Journal of Econometrics*, February 2016, *190* (2), 212–221.

Silva, J. M. C. Santos and Silvana Tenreyro, "The Log of Gravity," *The Review of Economics and Statistics*, November 2006, *88* (4), 641–658.

Simonovska, Ina and Michael E. Waugh, "The Elasticity of Trade: Estimates and Evidence," *Journal of International Economics*, 2014, *92* (1), 34–50.

Stock, James H. and Motohiro Yogo, "Testing for Weak Instruments in Linear IV Regression," in Donald W. K. Andrews and James H. Stock, eds., *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*, New York: Cambridge University Press, 2005, chapter 5, pp. 80–108.

Topalova, Petia and Amit Khandelwal, "Trade Liberalization and Firm Productivity: The Case of India," *The Review of Economics and Statistics*, August 2011, *93* (3), 995–1009.

U.S. Bureau of Labor Statistics, "Consumer Price Index: Relative Importance of Components in the Consumer Price Indexes, December 2018," U.S. Department of Labor, Washington, DC December 2018.

U.S. Census Bureau, "2017 Economic Census: Manufacturing," U.S. Department of Commerce, Washington, DC 2017. Machine-readable database.

_ , "Annual Survey of Manufactures," U.S. Department of Commerce, Washington, DC 2019. 2013, 2015, and 2019 survey years; machine-readable database.

_ , "USA Trade Online: State Merchandise Trade Statistics," U.S. Department of Commerce, Washington, DC 2020. State imports and exports by NAICS commodity.

— **and Bureau of Transportation Statistics**, "2017 Commodity Flow Survey," U.S. Department of Commerce and U.S. Department of Transportation, Washington, DC 2017. Machine-readable database.

U.S. Department of Commerce, Bureau of Industry and Security, Office of Technology Evaluation, "The Effect of Imports of Aluminum on the National Security: An Investigation Conducted under Section 232 of the Trade Expansion Act of 1962, as Amended," Report, U.S. Department of Commerce, Washington, DC January 2018. Report completed January 17, 2018; released February 16, 2018.

— , "The Effect of Imports of Steel on the National Security: An Investigation Conducted under Section 232 of the Trade Expansion Act of 1962, as Amended," Report, U.S. Department of Commerce, Washington, DC January 2018. Report completed January 11, 2018; released February 16, 2018.

U.S. International Trade Commission, "Crystalline Silicon Photovoltaic Cells (Whether or Not Partially or Fully Assembled into Other Products): Investigation No. TA-201-075," USITC Publication 4739, U.S. International Trade Commission, Washington, DC November 2017. Report to the President transmitted November 13, 2017.

— , "Large Residential Washers: Investigation No. TA-201-076," USITC Publication 4745, U.S. International Trade Commission, Washington, DC December 2017. Report to the President transmitted December 4, 2017.

A Theory Appendix

A.1 Derivation of Change in Price Index

The CES price index in state-sector ik is

$$P_{ik} = \left[(p_{ik}^f)^{1-\sigma} + \sum_{s \in \mathcal{S}_{ik} \setminus \{f\}} (p_{ik}^s)^{1-\sigma} \right]^{\frac{1}{1-\sigma}},$$

where p_{ik}^f is the delivered price of the foreign variety, p_{ik}^s is the price of variety s, and $\mathcal{S}_{ik}$ is the set of available varieties in state-sector ik.

Taking logs,

$$\log P_{ik} = \frac{1}{1-\sigma} \log \left[(p_{ik}^f)^{1-\sigma} + \sum_{s \in \mathcal{S}_{ik} \setminus \{f\}} (p_{ik}^s)^{1-\sigma} \right].$$

Differentiating with respect to the foreign price, holding the prices of all other varieties fixed, gives

$$d \log P_{ik} = \frac{(p_{ik}^f)^{1-\sigma}}{(p_{ik}^f)^{1-\sigma} + \sum_{s \in \mathcal{S}_{ik} \setminus \{f\}} (p_{ik}^s)^{1-\sigma}} d \log p_{ik}^f.$$

Since

$$P_{ik}^{1-\sigma} = (p_{ik}^f)^{1-\sigma} + \sum_{s \in \mathcal{S}_{ik} \setminus \{f\}} (p_{ik}^s)^{1-\sigma},$$

this can be written as

$$d \log P_{ik} = \frac{(p_{ik}^f)^{1-\sigma}}{P_{ik}^{1-\sigma}} d \log p_{ik}^f.$$

Define the foreign import share in state-sector ik as

$$\lambda_{ik}^f \equiv \frac{M_{ik}}{E_{ik}} = \frac{(p_{ik}^f)^{1-\sigma}}{P_{ik}^{1-\sigma}}.$$

Then

$$d \log P_{ik} = \lambda_{ik}^f d \log p_{ik}^f.$$

Since $p_{ik}^f = p_k^f \tau_{ik}^f (1 + t_k)$, holding the foreign producer price and non-tariff trade costs fixed implies

$$d \log p_{ik}^f = d \log(1 + t_k).$$

Therefore,

$$d \log P_{ik} = \lambda_{ik}^f d \log(1 + t_k).$$

A.2 Derivation of Spatial Spillover Term

Consider a gravity equation for domestic shipments from origin state i to destination state j in sector k:

$$D_{ijk} = (p_{ik}\tau_{ij})^{1-\sigma} P_{jk}^{\sigma-1} E_{jk}$$

where p_{ik} is the producer price in state i, τ_{ij} is the bilateral trade cost, P_{jk} is the destination price index, E_{jk} is total expenditure, and $\sigma > 1$ is the elasticity of substitution. Taking logs:

$$\log D_{ijk} = (1-\sigma)\log(p_{ik}\tau_{ij}) + (\sigma - 1)\log P_{jk} + \log E_{jk}$$

Differentiating with respect to the tariff, holding origin prices p_{ik}, trade costs τ_{ij}, and destination expenditure E_{jk} fixed:

$$d\log D_{ijk} = (\sigma - 1)\, d\log P_{jk}$$

A tariff on foreign imports raises the destination price index in proportion to the foreign import share:

$$d\log P_{jk} = \lambda_{jk}^{f}\, d\log(1 + t_k)$$

where $\lambda_{jk}^{f} = M_{jk,0}/E_{jk,0}$ is the initial foreign import share in destination market j, k. Substituting and converting to levels:

$$dD_{ijk} = D_{ijk,0}\,(\sigma - 1)\,\lambda_{jk}^{f}\, d\log(1 + t_k)$$

Total output of state i in sector k is the sum of its shipments to all destinations: $Y_{ik} = \sum_j D_{ijk}$. The spillover component of the output change—excluding own-state demand—is:

$$dY_{ik}^{spill} = \sum_{j \neq i} dD_{ijk} = (\sigma - 1)\, d\log(1 + t_k) \sum_{j \neq i} D_{ijk,0}\,\lambda_{jk}^{f}$$

Now relate this to import changes. Foreign imports into destination j, k change by:

$$dM_{jk} = (1 - \sigma)\, M_{jk,0}\, d\log(1 + t_k)$$

Using $D_{jk,0}^{dom} = E_{jk,0} - M_{jk,0}$ and substituting, the bilateral shipment gain can be rewritten as:

$$dD_{ijk} = -\frac{D_{ijk,0}}{D_{jk,0}^{dom}}\, dM_{jk}$$

Summing over $j \neq i$:

$$dY_{ik}^{spill} = -\sum_{j \neq i} \frac{D_{ijk,0}}{\sum_{\ell} D_{\ell jk,0}} \, dM_{jk}$$

This motivates the empirical spillover variable:

$$S_{ik}^{M} = \sum_{j \neq i} \frac{D_{ijk,0}}{\sum_{\ell} D_{\ell jk,0}} \, \Delta M_{jk}$$

The gravity framework predicts a coefficient of -1 on S_{ik}^{M}: a \$1 reduction in imports into destination state j raises shipments from state i in proportion to i's initial domestic market share in j.

A.3 Migration Model

Individual i, initially residing in state o and working in sector k, chooses a destination state d and destination sector k'. Utility from choosing (d, k') is

$$U_{iodkk'} = \alpha \log w_{dk'} - \alpha \log P_d + Z_d + Z_{k'} - C_{od} - C_{kk'} + e_{iodkk'}.$$

Assuming $e_{iodkk'}$ is distributed i.i.d. Type-I Extreme Value, the log choice probability is (McFadden 1978):

$$\log \frac{f_{odkk'}}{L_{ok}} = \alpha \log w_{dk'} - \alpha \log P_d + Z_d + Z_{k'} - C_{od} - C_{kk'} - \Phi_{ok},$$

where $f_{odkk'}$ is the flow from origin state-sector (o, k) to destination state-sector (d, k'), L_{ok} is employment in the origin state-sector, and

$$\Phi_{ok} = \log \left[\sum_{d', k''} \exp \left(\alpha \log w_{d'k''} - \alpha \log P_{d'} + Z_{d'} + Z_{k''} - C_{od'} - C_{kk''} \right) \right].$$

The term Φ_{ok} is the inclusive value for workers initially in (o, k), summarizing the attractiveness of all destination-sector options available to them.

Equivalently,

$$f_{odkk'} = L_{ok} \exp \left(\alpha \log w_{dk'} - \alpha \log P_d + Z_d + Z_{k'} - C_{od} - C_{kk'} - \Phi_{ok} \right).$$

The ACS reports current state, state of residence one year ago, and current industry, but not previous industry. I therefore aggregate over origin sectors k. Define

$$f_{odk'} \equiv \sum_k f_{odkk'}.$$

Substituting the expression above and summing over k,

$$f_{odk'} = \exp\left(\alpha \log w_{dk'} - \alpha \log P_d + Z_d + Z_{k'} - C_{od}\right) \sum_k L_{ok} \exp\left(-C_{kk'} - \Phi_{ok}\right).$$

Define

$$\Lambda_{ok'} \equiv \sum_k L_{ok} \exp\left(-C_{kk'} - \Phi_{ok}\right).$$

Then

$$f_{odk'} = \exp\left(\alpha \log w_{dk'} - \alpha \log P_d + Z_d + Z_{k'} - C_{od}\right) \Lambda_{ok'}.$$

Taking logs,

$$\log f_{odk'} = \alpha \log w_{dk'} - \alpha \log P_d + Z_d + Z_{k'} - C_{od} + \log \Lambda_{ok'}.$$

First-differencing gives

$$\Delta \log f_{odk'} = \alpha \Delta \log w_{dk'} + \underbrace{\left(-\alpha \Delta \log P_d + \Delta Z_d - \Delta C_{od}\right)}_{\Delta \delta_{od}} + \underbrace{\left(\Delta Z_{k'} + \Delta \log \Lambda_{ok'}\right)}_{\Delta \delta_{ok'}}.$$

I proxy destination state-sector wage growth with the import-protection instrument $Z^M_{dk'}$:

$$\Delta \log w_{dk'} = \gamma Z^M_{dk'} + \zeta_{odk'},$$

where $\zeta_{odk'}$ is the component of wage growth orthogonal to the tariff-induced labor demand shock. Substituting this into the first-differenced migration equation gives

$$\Delta \log f_{odk'} = \beta Z^M_{dk'} + \Delta \delta_{ok'} + \Delta \delta_{od} + \varepsilon_{odk'},$$

where $\beta \equiv \alpha\gamma$.

The fixed effect $\Delta \delta_{ok'}$ absorbs changes in the effective supply of workers from origin state o to sector k', including the attractiveness of outside options $\Delta \log \Lambda_{ok'}$ and sector-specific amenities. The fixed effect $\Delta \delta_{od}$ absorbs changes in origin-destination migration costs and destination-state shocks.

This now matches the specification in Section 4.2.

A.4 Derivation of Semi-Elasticity of Revenue to Tariff

Tariff revenue is

$$R = tpQ,$$

where p is the pre-tariff import price, t is the ad valorem tariff rate, and Q is the quantity of imports. Let $q \equiv p(1+t)$ denote the tariff-inclusive import price. I derive the semi-elasticity of revenue with respect to $\log(1+t)$.

Differentiating revenue with respect to $1+t$ gives

$$\frac{\partial R}{\partial(1+t)} = pQ + tp\frac{\partial Q}{\partial(1+t)} + tQ\frac{\partial p}{\partial(1+t)}.$$

Multiplying both sides by $1+t$ gives

$$\frac{\partial R}{\partial \log(1+t)} = (1+t)pQ + tpQ\frac{\partial \log Q}{\partial \log(1+t)} + tpQ\frac{1+t}{p}\frac{\partial p}{\partial(1+t)}.$$

For the import-demand term, the chain rule implies

$$\frac{\partial \log Q}{\partial \log(1+t)} = \frac{\partial \log Q}{\partial \log q}\frac{\partial \log q}{\partial \log(1+t)} = \rho\epsilon_t^D,$$

where

$$\epsilon_t^D \equiv \frac{\partial \log Q}{\partial \log q} \quad \text{and} \quad \rho \equiv \frac{\partial \log q}{\partial \log(1+t)}$$

are the import-demand elasticity and tariff pass-through rate, respectively.

For the price term, since $q = p(1+t)$,

$$\log q = \log p + \log(1+t).$$

Therefore,

$$\rho = \frac{\partial \log q}{\partial \log(1+t)} = \frac{\partial \log p}{\partial \log(1+t)} + 1,$$

which implies

$$\frac{1+t}{p}\frac{\partial p}{\partial(1+t)} = \rho - 1.$$

Substituting these expressions into the revenue derivative gives

$$\frac{\partial R}{\partial \log(1+t)} = (1+t)pQ + tpQ\rho\epsilon_t^D + tpQ(\rho - 1).$$

Collecting terms:

$$\frac{\partial R}{\partial \log(1+t)} = pQ\left[(1+t) + t\rho\epsilon_t^D + t(\rho-1)\right],$$

so

$$\frac{\partial R}{\partial \log(1+t)} = pQ\left[1 + t\rho(\epsilon_t^D + 1)\right].$$

B Econometrics Appendix

B.1 Sufficiency of Identification Condition

The IV orthogonality condition needed for consistent estimation is:

$$\mathbb{E}\left[M_{ik,2017}\Delta\log(1+t_k)\Delta e_{ik} \mid \Delta\delta_i, \Delta\delta_k, G_{ik}\right] = 0. \tag{10}$$

I now show that the following condition is sufficient:

$$\mathbb{E}\left[\Delta\log(1+t_k)\Delta e_{ik} \mid M_{ik,2017}, \Delta\delta_i, \Delta\delta_k, G_{ik}\right] = 0. \tag{11}$$

By the law of iterated expectations,

$$\begin{aligned}
&\mathbb{E}\left[M_{ik,2017}\Delta\log(1+t_k)\Delta e_{ik} \mid \Delta\delta_i, \Delta\delta_k, G_{ik}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[M_{ik,2017}\Delta\log(1+t_k)\Delta e_{ik} \mid M_{ik,2017}, \Delta\delta_i, \Delta\delta_k, G_{ik}\right] \mid \Delta\delta_i, \Delta\delta_k, G_{ik}\right] \\
&= \mathbb{E}\left[M_{ik,2017}\mathbb{E}\left[\Delta\log(1+t_k)\Delta e_{ik} \mid M_{ik,2017}, \Delta\delta_i, \Delta\delta_k, G_{ik}\right] \mid \Delta\delta_i, \Delta\delta_k, G_{ik}\right] \\
&= 0,
\end{aligned} \tag{12}$$

where the final equality follows from Equation 11. This proves Equation 10.

B.2 Comparison With Shift-Share Design

This appendix formally contrasts the shift-share design and the design in my paper. For ease of exposition, I omit controls from this discussion.

A shift-share design is interested in some outcome at the regional level, e.g. state or county level. For example, a researcher is interested in how output Y_i in a state i is affected by its import exposure. A standard shift-share in this literature is defined as:

$$Z_i^{SS} = \sum_k s_{ik,0}\Delta g_k$$

where $s_{ik,0}$ is the baseline share of sector k employment in state i, and Δg_k is an aggregate shock to sector k. For example, Δg_k could be changes in nationwide sector tariffs, as in this paper; in Autor, Dorn and Hanson (2013), Δg_k is growth in Chinese imports in sector k. The shift-share regression is:

$$\Delta Y_i = \beta Z_i^{SS} + \Delta e_i$$

By contrast, the specification in my paper is:

$$\Delta Y_{ik} = \beta \Delta M_{ik} + \Delta\delta_i + \Delta\delta_k + \Delta e_{ik}$$

where Y_{ik} is manufacturing output in state i and sector k, ΔM_{ik} is the change in imports in state i in sector k, and $\Delta\delta_i$ and $\Delta\delta_k$ are state and sector fixed effects.

I instrument ΔM_{ik} with

$$Z_{ik}^M = M_{ik,0}\Delta g_k$$

where $M_{ik,0}$ is baseline imports in state-sector ik. In this paper, $\Delta g_k = \Delta \log(1 + t_k)$ where t_k is the nationwide tariff in sector k.

Shift-share designs compare regions with different industrial compositions after a nationwide sector-level shock, whereas my design compares the same industry across regions with different baseline import exposure. Put differently, the shift-share regression studies outcome changes at the regional level i, whereas my regression studies at the regional-sector level ik. This allows me to include sector fixed effects and compare across regions within the same sector.

These two approaches rely on different identification assumptions. As shown in Section 2, the identification condition for this design is:

$$\mathbb{E}(\Delta g_k \cdot \Delta e_{ik} \mid M_{ik}, \Delta\delta_i, \Delta\delta_k) = 0$$

Exogenous Shocks Shift-share identification arguments typically rely on one of two views. I first consider the exogenous-shocks view of Borusyak, Hull and Jaravel (2022).

$$\mathbb{E}(\Delta g_k \cdot \Delta\bar{e}_k \cdot s_k) = 0$$

where $s_k = \sum_i s_{ik}$, and $\bar{e}_k = \frac{\sum_i s_{ik}e_i}{\sum_i s_{ik}}$

Shift-share designs requires the national shock Δg_k to be uncorrelated with *nationwide* sector-level shocks $\Delta\bar{e}_k$; my design instead requires the national shock Δg_k to be uncorrelated with *local* state-sector shocks Δe_{ik}. In this sense the identification condition is more granular.[25] In addition, my design allows for the inclusion of sector fixed effects $\Delta\delta_k$, unlike shift-share designs. Therefore, any sector-wide shock common to all regional units i, such as technological breakthroughs or federal policy changes, may confound shift-share designs, while being absorbed by sector fixed effects in my design.

Exogenous Shares Now I compare my design instead to the exogenous-shares view of Goldsmith-Pinkham, Sorkin and Swift (2020):

$$\mathbb{E}(s_{ik,0} \cdot \Delta e_i) = 0$$

The identification condition here requires that initial industrial composition is uncorrelated to future location shocks. In this sense, the exogenous-shares view is more granular than the exogenous-shocks view: identification comes from the interaction of a region-sector variable, $s_{ik,0}$, with a regional shock, Δe_i, rather than from the interaction of a sector-level shock, Δg_k, with a sector-level residual, $\Delta \bar{e}_k$. However, this shifts the identifying assumption onto the determinants of baseline industrial composition. In the trade context, this assumption is demanding: regions specializing in different industries may also be differentially exposed to non-trade shocks, such as immigration flows, automation, unionization drives, or region-specific industrial decline.

B.3 Sensitivity to Export Pre-Trends Using Rambachan and Roth (2023)

This appendix describes the implementation of the Rambachan and Roth (2023) sensitivity analysis used in Section 5. I apply the procedure to the reduced-form export-retaliation coefficient, since the 2015–2017 pre-trend appears in the reduced form.

Define the pre-period reduced-form coefficient π_X^{pre} from

$$Y_{ik,2017} - Y_{ik,2015} = \pi_X^{pre} Z_{ik}^X + \pi_M^{pre} Z_{ik}^M + \Delta\delta_i + \Delta\delta_k + \eta_{ik}^{pre}.$$

Similarly, define the post-period reduced-form coefficient π_X^{post} from

$$Y_{ik,2019} - Y_{ik,2017} = \pi_X^{post} Z_{ik}^X + \pi_M^{post} Z_{ik}^M + \Delta\delta_i + \Delta\delta_k + \eta_{ik}^{post}.$$

[25] In some contexts, where the nationwide shock Δg_k is a policy lever such as tariffs, this condition is arguably more plausible because violating it requires a more specific form of targeting. For example, policymakers targeting fast-growing sectors nationally would threaten shift-share designs, but would not threaten my design unless they targeted sectors based on differential trends across states within the same sector, which is a higher informational and cognitive burden on policymakers.

Following Rambachan and Roth (2023), decompose the export reduced-form coefficients as:

$$\begin{pmatrix} \pi_X^{pre} \\ \pi_X^{post} \end{pmatrix} = \begin{pmatrix} 0 \\ \rho_X \end{pmatrix} + \begin{pmatrix} \gamma^{pre} \\ \gamma^{post} \end{pmatrix},$$

where ρ_X is the causal reduced-form effect of the export-retaliation instrument on output, and γ^{pre} and γ^{post} are the pre- and post-period violations of parallel trends. Since export retaliation has not occurred in the pre-period, $\pi_X^{pre} = \gamma^{pre}$. I impose the relative-magnitudes restriction

$$|\gamma^{post}| \leq M|\gamma^{pre}|.$$

The sensitivity parameter M governs how large the post-period violation is allowed to be relative to the observed pre-period violation. Thus, $M = 0$ imposes no post-period violation, $M = 1$ allows the post-period violation to be as large as the observed pre-period violation, and $M = 2$ allows it to be twice as large. Because the restriction is imposed in absolute value, the post-period violation may have the opposite sign of the pre-period violation.

I implement this restriction using the estimated pair $(\hat{\pi}_X^{pre}, \hat{\pi}_X^{post})$ and its variance-covariance matrix. This yields a sensitivity confidence set for the causal reduced-form effect of the export-retaliation instrument on output. I then map this reduced-form confidence set into structural effects using the estimated first-stage matrix.

Let the first-stage matrix be

$$\Gamma = \begin{pmatrix} \alpha_{MM} & \alpha_{MX} \\ \alpha_{XM} & \alpha_{XX} \end{pmatrix},$$

where α_{MM} is the coefficient of Z_{ik}^M in the first stage for imports, α_{MX} is the coefficient of Z_{ik}^M in the first stage for exports, α_{XM} is the coefficient of Z_{ik}^X in the first stage for imports, and α_{XX} is the coefficient of Z_{ik}^X in the first stage for exports.

The reduced-form coefficients and structural coefficients are linked by

$$\begin{pmatrix} \pi_M^{post} \\ \pi_X^{post} \end{pmatrix} = \Gamma \begin{pmatrix} \beta \\ \theta \end{pmatrix}.$$

Therefore,

$$\begin{pmatrix} \beta \\ \theta \end{pmatrix} = \Gamma^{-1} \begin{pmatrix} \pi_M^{post} \\ \pi_X^{post} \end{pmatrix}.$$

I apply the Rambachan and Roth sensitivity procedure to the export-retaliation reduced form, since this is where the pre-trend appears. The import reduced form is not separately adjusted for pre-trends, since no import-protection pre-trend is detected in Table 1. However, because the structural coefficients are recovered jointly through the first-stage matrix, the implied import coefficient can still change even without an import-protection pre-trend. The resulting reduced-form confidence sets are transformed through the estimated first-stage matrix to obtain the structural sensitivity intervals reported in Table 11.